

PROVINCETOWN, MA, USA

WMO: 725073

Lat: 42.072N

Lon: 70.221W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64708

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-7.9	-17.6	0.8	-8.5	-16.1	0.9	-6.8	15.7	-1.8	14.7	-2.4	8.3	320	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	28.7	22.7	27.4	21.6	26.1	20.9	23.8	27.2	22.9	25.8	22.0	24.4	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.2	25.6	22.1	16.8	25.0	21.2	15.9	23.9	70.9	27.3	67.3	25.8	64.3	24.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.9	10.7	-13.4	31.8	2.5	1.7	-15.2	33.0	-16.7	34.0	-18.2	34.9	-20.0	36.1		
(5)			-14.2	25.4	2.5	1.2	-16.0	26.2	-17.4	27.0	-18.8	27.6	-20.6	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	0.3	0.3	3.1	7.6	12.8	17.9	21.8	21.1	17.6	12.5	7.4	3.3		
	DBStd	8.36	4.60	4.06	3.65	3.31	3.46	3.35	2.59	2.72	3.25	3.51	3.89	4.15		
	HDD10.0	1204	301	271	216	86	11	0	0	0	0	16	95	209		
	HDD18.3	3119	559	504	473	321	176	48	4	8	52	181	328	465		
	CDD10.0	1398	0	0	2	14	96	236	365	343	228	94	17	2		
	CDD18.3	273	0	0	0	0	3	34	111	93	30	2	0	0		
	CDH23.3	1092	0	0	0	3	22	132	496	368	69	2	0	0		
	CDH26.7	189	0	0	0	1	3	22	102	55	7	0	0	0		
(14) Wind	WSAvg	5.2	6.4	6.2	5.9	5.1	4.5	4.2	4.0	3.8	4.3	5.2	5.9	6.3		
(15) Precipitation	PrecAvg	1079	98	85	98	91	79	77	61	81	91	93	115	113		
	PrecMax	1461	327	232	252	240	206	194	135	186	245	230	239	226		
	PrecMin	582	28	20	21	23	15	13	14	2	24	32	30	33		
	PrecStd	229	73	43	53	46	45	51	34	49	54	51	57	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	11.4	15.8	20.2	26.4	29.1	31.4	29.9	27.4	22.6	17.5	14.1		
		MCWB	11.3	9.7	10.6	14.8	18.2	21.8	23.7	23.9	21.9	18.9	14.7	12.2		
	2%	DB	9.8	8.9	12.0	16.8	22.5	26.7	29.0	27.9	25.3	20.9	16.1	12.5		
		MCWB	8.9	7.4	9.6	11.8	16.8	20.5	22.6	22.4	20.7	17.8	13.8	11.1		
	5%	DB	7.9	7.4	9.7	14.0	20.1	24.7	27.5	27.1	23.6	18.9	14.1	11.3		
		MCWB	6.5	5.7	7.5	10.3	15.0	19.1	21.8	21.7	19.8	16.8	11.9	10.0		
	10%	DB	6.9	6.1	7.8	12.6	17.8	22.8	26.2	25.9	22.4	17.6	12.9	9.5		
		MCWB	5.2	4.1	5.6	9.5	13.9	18.1	21.1	20.9	19.3	14.9	10.7	8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	10.3	11.7	15.6	19.8	23.2	25.1	24.9	23.5	20.4	16.1	13.1		
		MCDB	11.8	10.9	13.0	18.9	23.8	27.6	30.2	28.4	26.6	21.8	17.0	13.8		
	2%	WB	8.8	7.8	9.5	13.2	17.8	21.6	23.9	23.8	21.9	18.5	14.4	11.6		
		MCDB	9.3	8.6	11.1	15.5	20.9	25.3	27.5	26.9	23.7	19.5	15.9	12.3		
	5%	WB	7.2	6.1	7.8	11.7	16.3	20.5	22.9	22.9	20.9	17.2	12.8	10.1		
		MCDB	7.9	7.3	8.9	13.4	19.2	23.3	26.1	25.4	22.7	18.4	14.0	11.0		
	10%	WB	5.2	4.3	6.0	10.1	14.8	19.3	22.1	21.9	19.7	15.6	11.4	8.2		
		MCDB	6.6	5.8	7.7	12.3	17.5	21.8	24.7	24.0	21.6	17.3	12.8	9.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.3	6.1	7.1	7.8	7.9	7.8	7.6	7.3	6.6	6.3	5.8		
		MCDBR	7.2	7.8	8.6	9.9	11.3	10.3	9.3	8.7	8.2	7.9	7.3	7.5		
	5% WB	MCWBR	7.3	7.3	6.6	6.3	6.3	5.4	4.4	4.5	5.1	5.9	6.8	8.0		
		MCWBR	7.4	8.1	8.1	8.6	9.8	8.9	8.1	7.5	6.9	7.2	7.0	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.312	0.330	0.371	0.406	0.458	0.480	0.439	0.376	0.343	0.317	0.303		
		taud	2.520	2.509	2.498	2.403	2.354	2.240	2.185	2.292	2.448	2.539	2.572	2.572		
	Ebn at Noon		847	899	919	898	871	823	799	822	857	851	829	822		
		Edh at Noon	71	84	96	113	122	137	143	125	100	81	68	63		
(44) All-Sky Solar Radiation	RadAvg		1.67	2.57	3.63	4.74	5.36	5.76	6.00	5.40	4.35	2.93	1.92	1.41		
	RadStd		0.13	0.24	0.27	0.51	0.54	0.52	0.35	0.24	0.29	0.24	0.15	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+43

Nomenclature: See separate page